Digital Technologies for Precision Farming

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ABSTRACT

Precision farming is a technique that uses modern-day technologies. In this paper, we propose AVANURMI, an automated soil moisture control system using Time domain reflectometry(TDR) integrated with sound sensing controlled via ESP32, and uses Artificial Intelligence and machine learning to predict and/or show the data in a user-friendly application such as Krishi mapper.

INTRODUCTION

Precision Farming refers to a method of farming where technologies such as Remote sensing(RS), Drone technologies, Internet of Things (IOT), robotics, Geographic Information System(GIS), Artificial Intelligence, and Machine Learning Models are used to make agriculture more efficient use of resources and decrease wastage. Here, we can utilize the Global Positioning System (GPS) to collect data about specific locations and compare it with data from previous years to predict the future. Although these are some of the positive aspects, challenges such as unpredictable climate change, pest attacks, and sudden natural calamities remain ongoing issues.

REMOTE SENSING AND GIS

The use of Remote Sensing and GIS is quite popular in the field of precision farming. Sensors such as LIDAR, Thermal, and satellite imagery are being used in this field. Additionally, the data from these can be used to see the current status of the field and also predict the yield of the land. Here, Monteith's crop modelling equation, derived from Monteith (1972,1977), can be used.

Monteith's crop yield modelling equation [1]

$$\boxed{YEILD = APAR \times \varepsilon \times HI}$$

WHERE:

APAR=total amount of photosynthetically active radiation(PAR) absorbed by a canopy through the growing season

ε = the light use efficiency in units of g biomass MJ^{-1}

HI= the harvest index or ratio of grain mass to above-ground biomass.

But this equation comes with a major problem, as the reliability of the prediction in a highly managed system is unknown[1].

Other soil measurement sensors, like the soil moisture sensor,

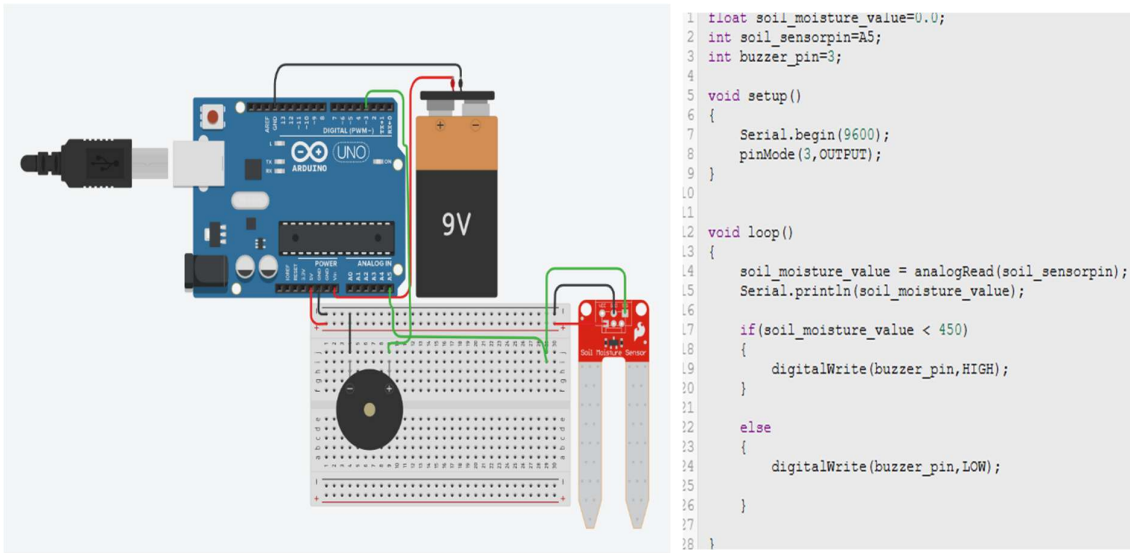


Fig 1. A simple circuit of a capacitive soil moisture sensor triggering system using Arduino UNO (simulation made in tinkercad.com)

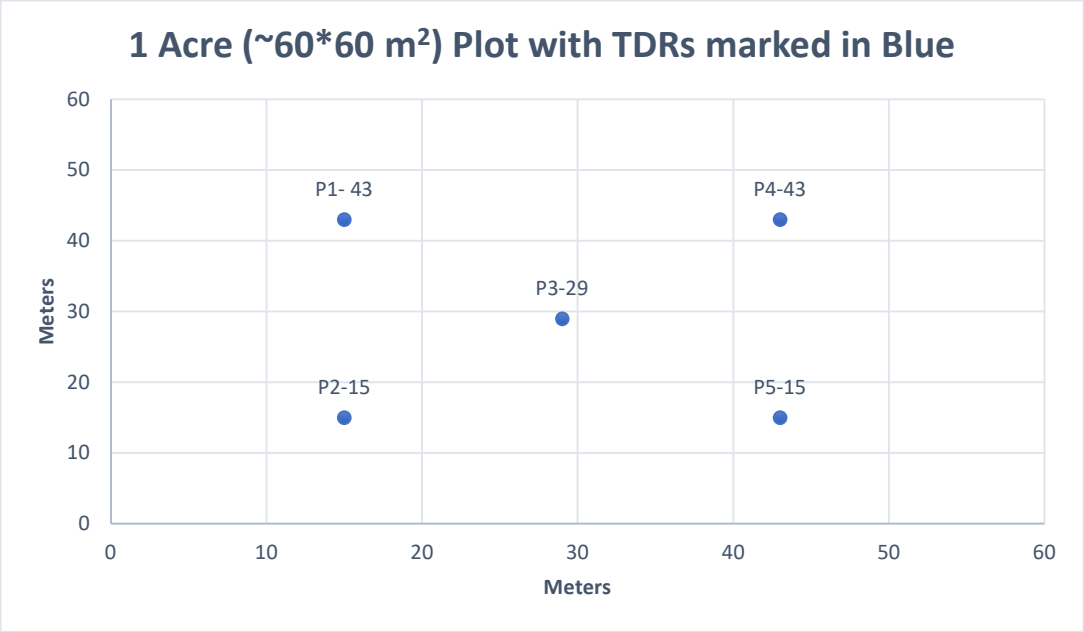
This can be used as a trigger system. Whenever the moisture level is dry

(~ 0-400) To trigger a sprinkler system automatically, but the capacitive soil moisture sensor given in Fig.1 has a moderate accuracy of 3%-5% a more reasonable approach for this will be to go with a Time Domain reflectometry(TDR) sensor, where an electromagnetic pulse is sent into the soil and the time taken by the pulse to reflect is noted. As water has a higher dielectric constant than dry soil or air, the longer travel time of the electromagnetic pulse indicates higher volumetric water content. TDR systems are more accurate with an error rate of up to ±3% [9], but these TDRs are not cheap. However, in the long term, this system will require minimal maintenance and can operate for up to 20 years. How many of these systems should be kept per acre to get a reasonable standard of accuracy? To create a complete automated system for the land?.[10]

Let 1point = 1 location in the acre where we insert the TDR.[10] We often would use data from the place with different depths (eg,10cm,20cm,30 cm)

The collective data of these different depths will be 1 point.

[10] To achieve a reasonable amount of accuracy, we will need at least a 5-point system. 1acre = 4046.86 m^2 i.e, ~4047 m^2 . Therefore, 4047/5 = 809.4m^2 or~28.44 m. Therefore, a 5-point system will need to be placed with a special distance of ~28.44m between each other. The picture below (Fig.2) shows a scatter plot with the 5-point system marked from p1-p5 along with their respective Y values.



Furthermore, Technology like sound sensing can be integrated with these kinds of systems, and data can be fed through apps like "Krishi mapper" (by The Ministry of Agriculture and Farmer Welfare(MOA&FW), Government of India) so that the farmers can always be in touch with their field.

Integrating sound sensing with soil moisture sensors

Some pests like cicadas, grasshoppers, and crickets are known for their unique sound, particularly by male cicadas, to attract female cicadas. These sounds create a unique pattern of frequencies.

These frequencies can be used to identify pest attacks in specific areas of the field. Therefore, it helps us to take the necessary steps to control pest activity in fields.

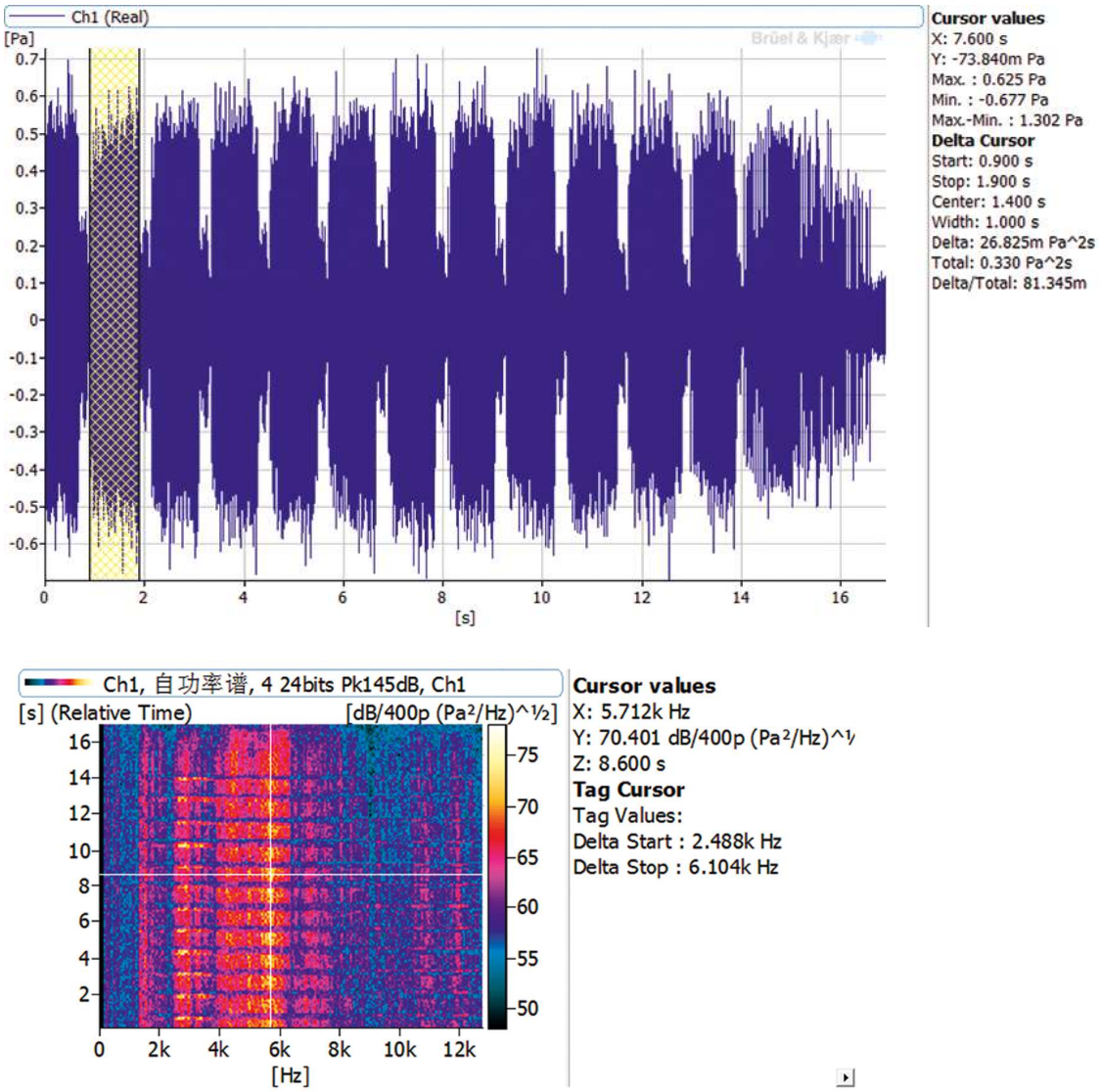


Fig 3. The oscillogram and spectrogram of a cicada's chirp sound. (Source: Why do Circadas Chirp? The Sound of Summer – Liangwei Shen. Brüel & Kjær China [8])

In the Fig 3, given above, the frequency ranges from 20Hz-12.8kHz, with the dominant frequency around 2.5kHz and 6.1KHz. This kind of data for different insects can be taken, as shown in Fig1 we can connect this TDR soil moisture sensor system with a sound sensing system to an industrial Controller, we call

this system “AVANURMI” (meaning soil and waves in Sanskrit), not necessarily an Arduino, an ESP32 (industrial standard) could also be used.

Using the Internet of Things (IOT) in AVANURMI

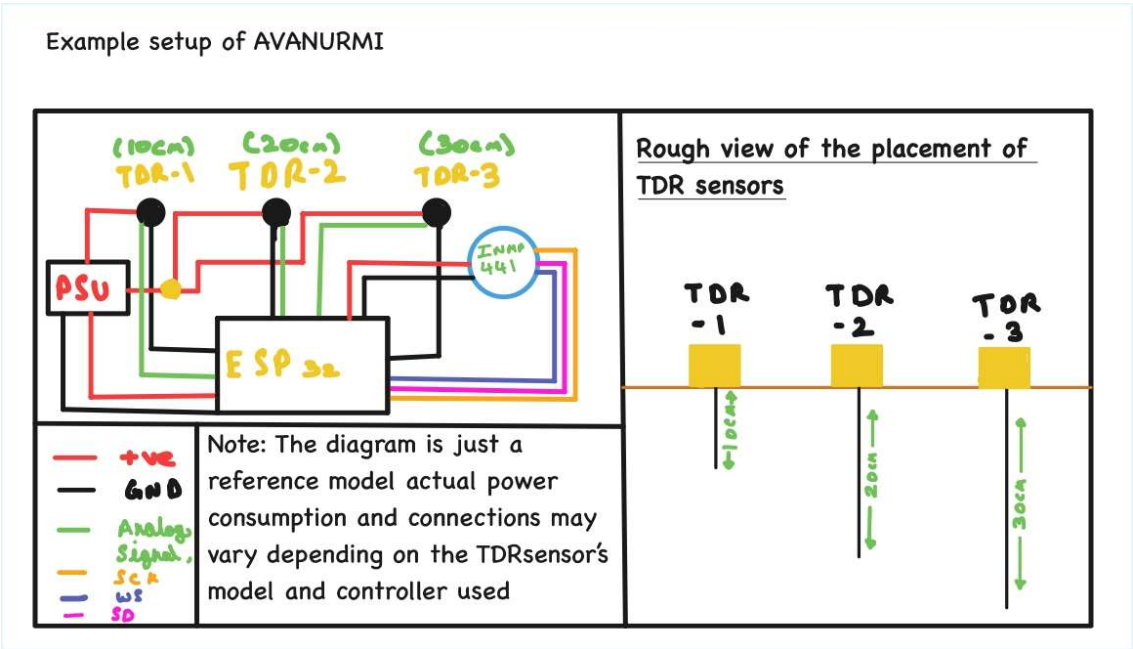


Fig 4. A reference model of the Avanurmi system.

In the figure given above, we can see the ideal setup of the Avanurmi setup, where an ESP32 is used to collect data from the TDR soil moisture sensors and the INMP 441 microphone for sound sensing. These data can be sent in real time using the ESP32’s WIFI/Bluetooth.

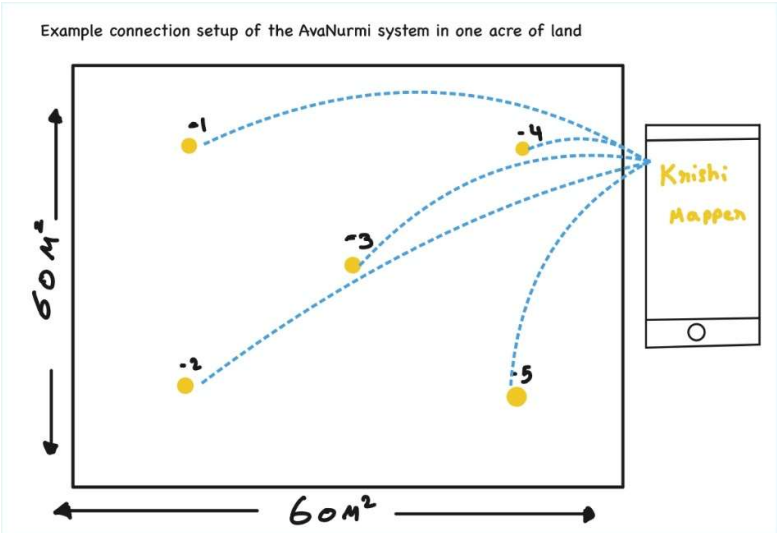


Fig 4a. A diagrammatic representation of A 5-point Avanurmi system

Another approach for sending this data could be by using the Internet. The data received from the TDR soil moisture sensors and the INMP 441 can be sent directly to the cloud from the ESP32. As basic as it sounds, the ESP32 in the field will be connected to a stable Wifi network through which the ESP32 will send the data to cloud operators such as AWS IoT Core, Azure IOT hub, or a more cost-efficient approach would be to go with ThingSpeak's platform to visualize the image. But this requires farmers to go through some technical training, which can be hard to implement. So, what can we do about this? This is where artificial intelligence and machine learning algorithms help.

Integrating Artificial Intelligence and Machine Learning to automate data processing in Avanurmi

After storing the data in the cloud servers, this data can now be cleaned and given to AI models such as CosmoAIgraph by Microsoft [11], or Amazon Neptune's graph RAG model can be used to automate this data, and the visualized version of this data can be shown in Krishi mapper-like apps to know the status of the field.

Previously collected data on soil moisture can be used to calculate the regression of the field during different seasons and chemicals, such as calcium chloride, silica gel, or to reduce soil moisture during flood seasons use of quicklime, lime kiln dust, and calciment can be used[12]. For this regression algorithm, such as random forest, decision trees can be used.

Prospects of Avanurmi

- Soil moisture and pest attack information on the mobile phone 24/7.
- Soil moisture information plays a crucial role in water management and water resource planning[12], for which Avanurmi could be useful.
- Agricultural maintenance for large-scale lands will be less burdensome.
- Avanurmi trigger system could be connected to a separate wireless sprinkler system, which gets triggered for a specific area if that area's closest point system reads low moisture levels.
- *"A new study warns that global declines in soil moisture in the 21st century could mark a "permanent" shift in the world's water cycle*."[13]. The Avanurmi system can help in solving this problem by making the stats available anytime, anywhere.
- One step towards Digital India.

Difficulties in implementing Avanurmi

- High cost – not suitable for small farming fields.
- A stable internet connection is needed, which may not be available in farm lands.
- Farmers/people may need to undergo some training using the Krishi mapper app to see the visualized data.
- Setup of Avanurmi may be complex, but in the long run, the maintenance is relatively less if installed properly and protectively.

Conclusion

The Avanurmi system, although it is expensive, is a long-term benefit, especially for farms with a larger area (>=1acre). As artificial intelligence gets more sophisticated features, like recommendation of appropriate chemical treatments for a specific part of the field, it can be mentioned in the Krishi mapper app, along with the data given by Avanurmi. This system encourages a modern way of precision farming and contributes to India's transition towards digital empowerment in agriculture.

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